Topic 5: Communication and the internet

Computer networks and the internet are now ubiquitous. Many computer applications in use today would not be possible without networks. Students should understand the key principles behind the organisation and of computer networks. Ideally, they should be able to experiment by setting up a simple network

Students should be given the opportunity to gain practical experience of creating web pages.

Subject content	Students should:
5.1 Networks	5.1.1 Understand the advantages and disadvantages of connecting computers in a network
	5.1.2 Understand the different types of networks [LAN, WAN, PAN, VPN]
	5.1.3 Understand the network media [copper cable, fibre optic cable, wireless]
	5.1.4 Understand that network data speeds are measured in bits per second [Mbps, Gbps]
	5.1.5 Understand the role of and need for standard network protocols
	5.1.6 Understand that data is transmitted over networks using packets [TCP/IP]
	5.1.7 Understand that data transmitted over a network is subject to errors [interference, power spikes, interception, cross talk]
	5.1.8 Be familiar with the use of check sums to detect errors in data transmission
	5.1.9 Understand the concept of and need for network addressing and host names [MAC addresses]
	5.1.10 Understand network topologies [bus, ring, star, mesh]